

Architecting the Credential Attack Suite

A Safe, Ethical Framework for Vulnerability Assessment



DEVELOPER :: Jay Thakkar

CONTEXT :: Internship Project

REPOSITORY :: Password_Cracking_Credential_Attack_Suite

The Mandate: Ethical Simulation of Risk

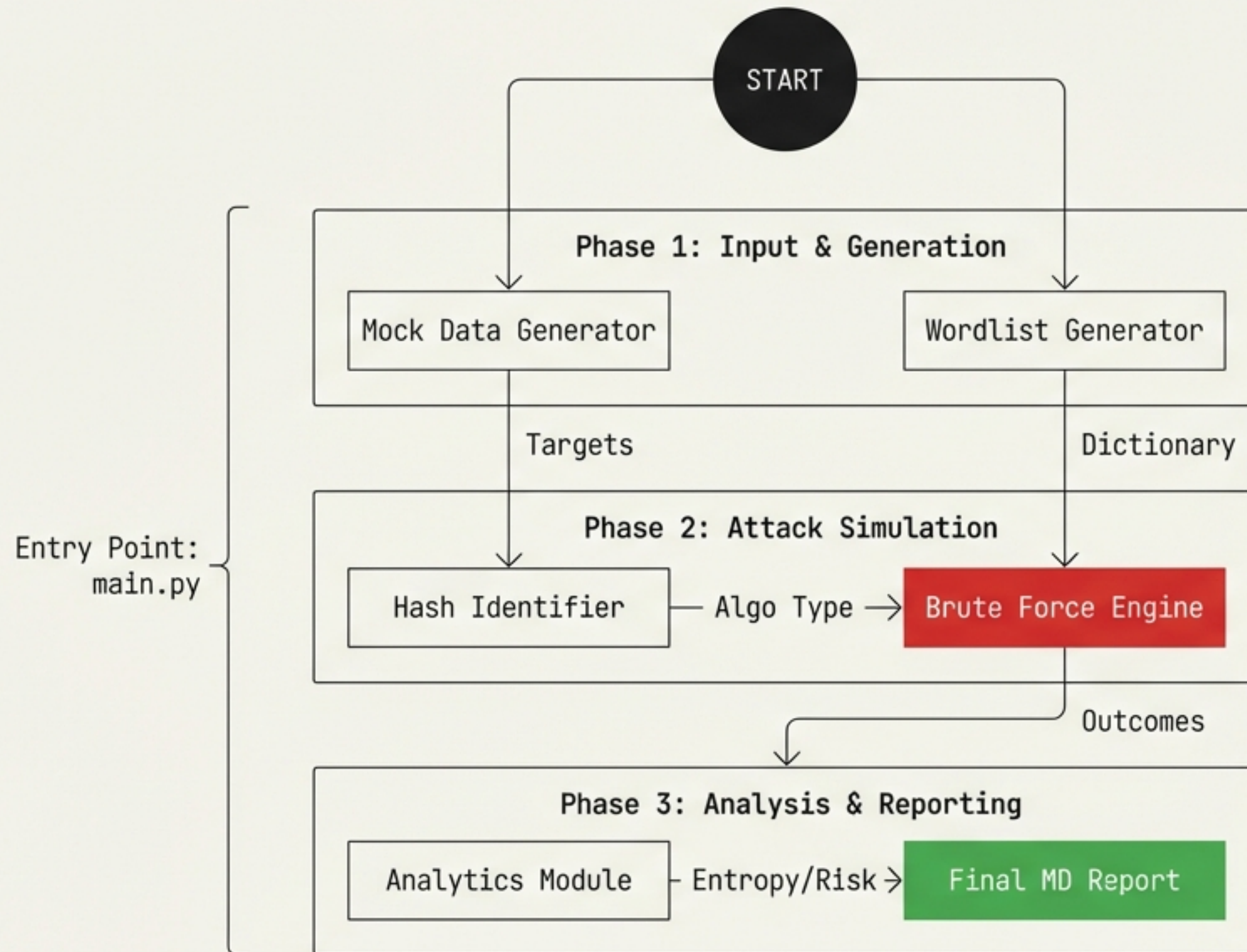
THE CHALLENGE

Simulating the lifecycle of a modern password audit without interacting with real user data.

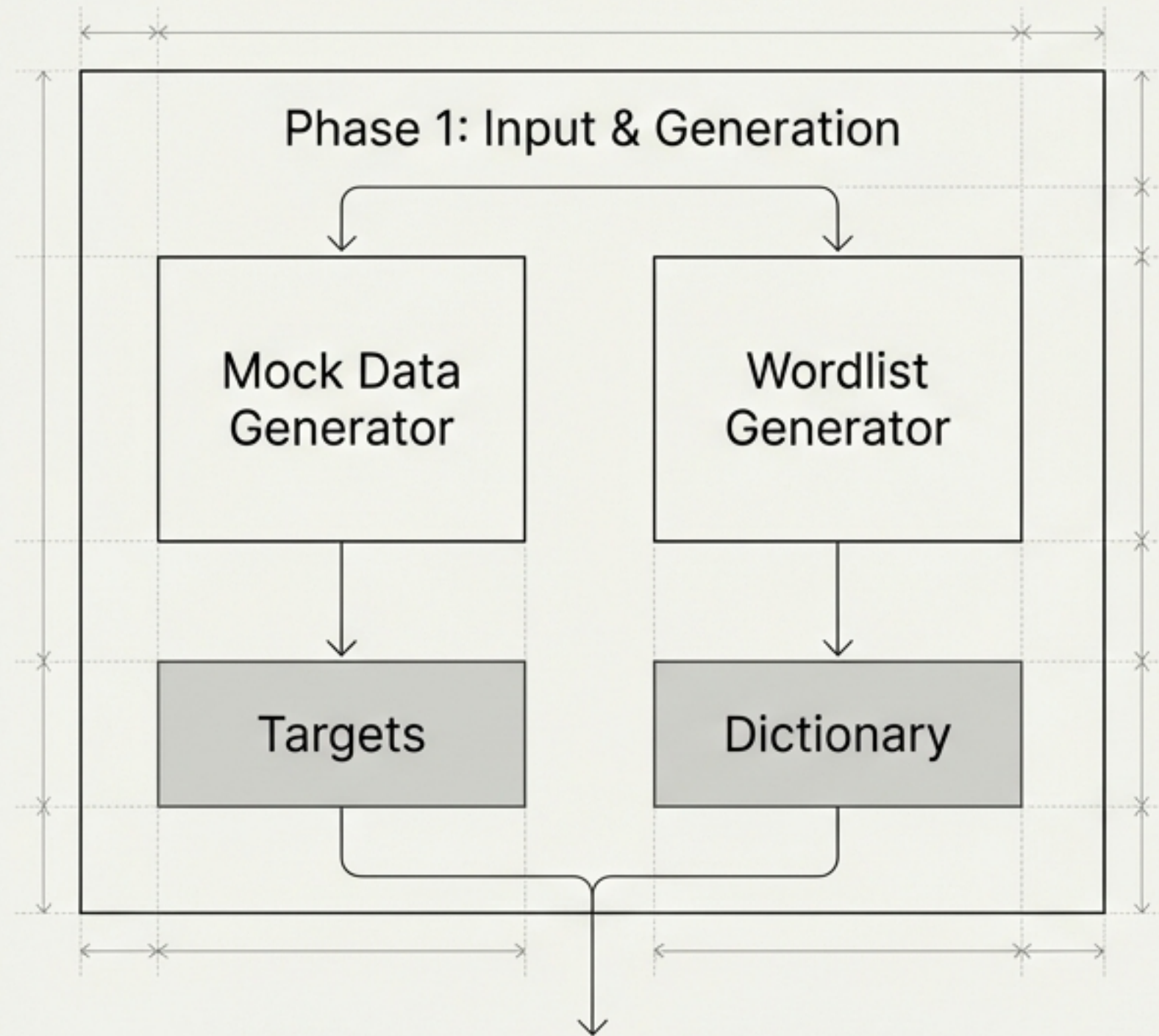
THE OBJECTIVE

- ✓ **Safety:** Operation entirely within a simulation environment using mock data.
- ✓ **Transition:** Shift from “Red Team” vulnerability assessment to “Blue Team” defense strategy.
- ✓ **Outcome:** Producing professional audit reports with mitigation recommendations.

The Pipeline Architecture

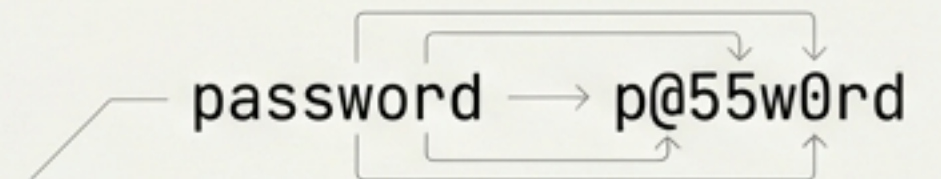


Phase 1: Engineering the Attack Surface



Module A: Mock Data Generator

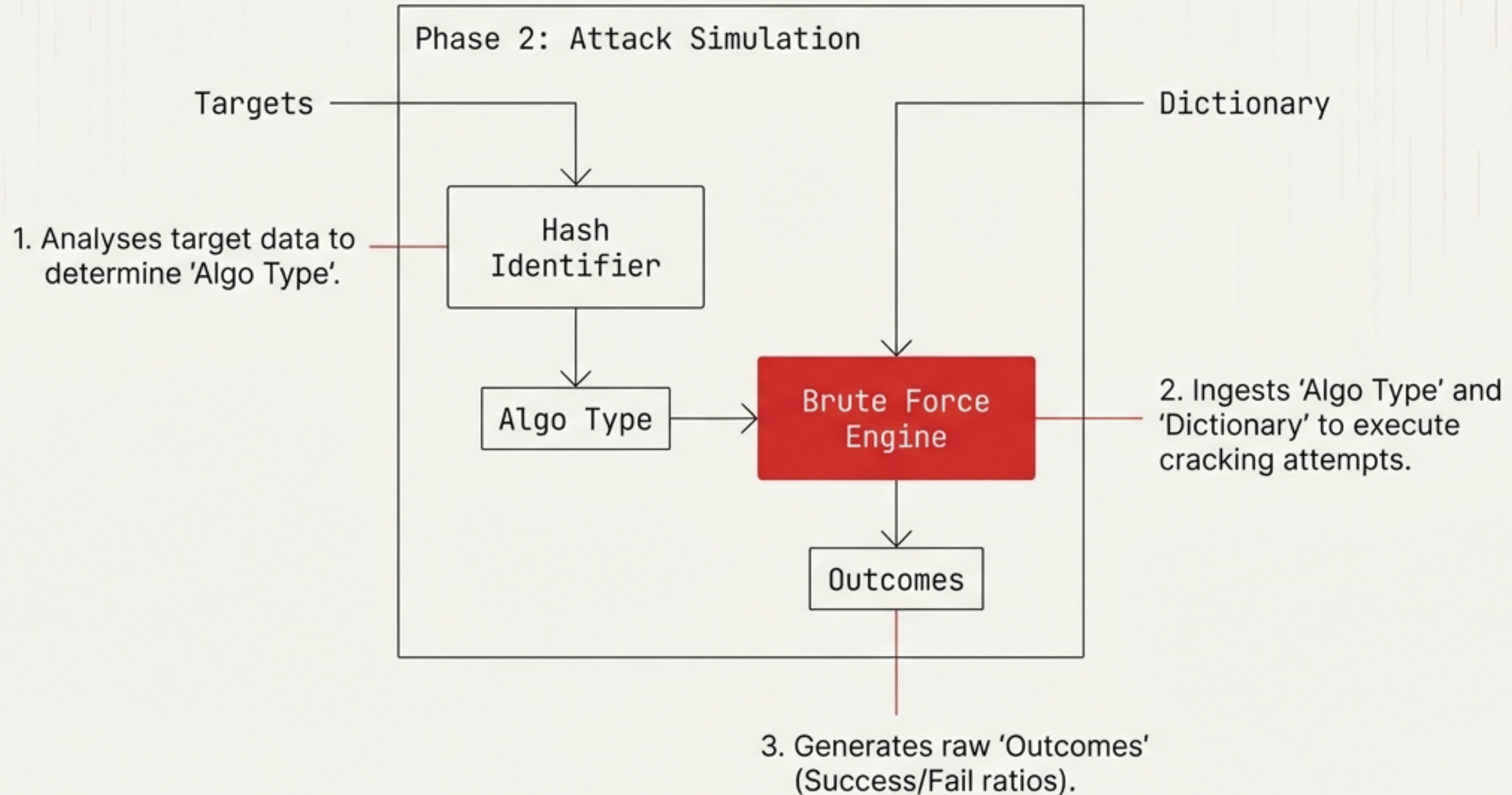
- Simulates storage formats of real operating systems.
- Mimics Linux /etc/shadow and Windows SAM formats.
- Output: Generates 'Targets' for the attack engine.



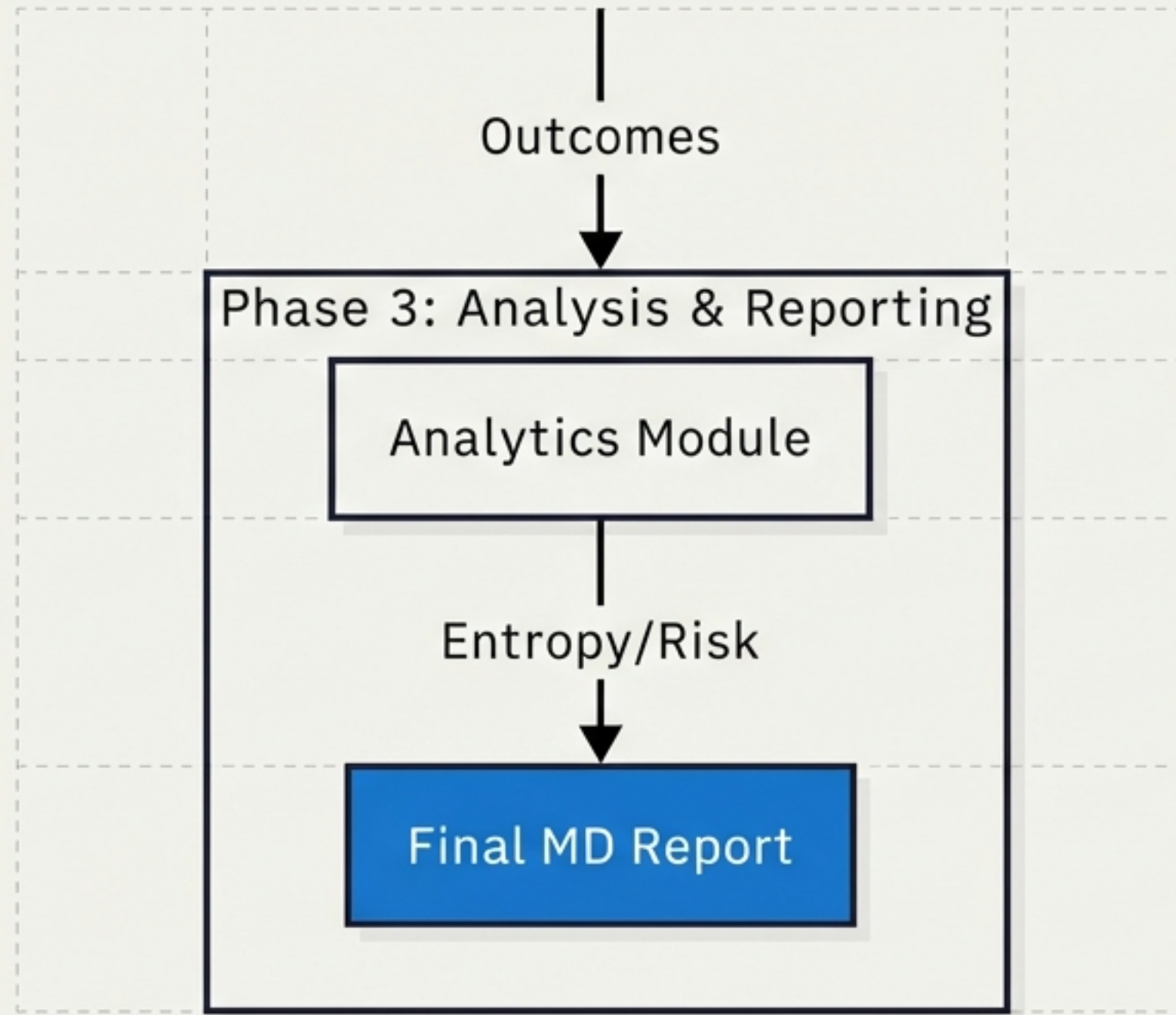
Module B: Dictionary Generator

- Engineers custom wordlists for testing.
- Uses mutation patterns (leet-speak) to enhance standard dictionaries.
- Output: Generates a "Dictionary".

Phase 2: The Resilience Simulation Engine



Phase 3: From Raw Data to Security Intelligence



INPUT :: Raw Attack Outcomes



PROCESS :: Analytics Module



METRICS ::
Password Entropy &
Estimated Cracking Time



OUTPUT :: Final_Security_Report.md

Technical Implementation & Stack

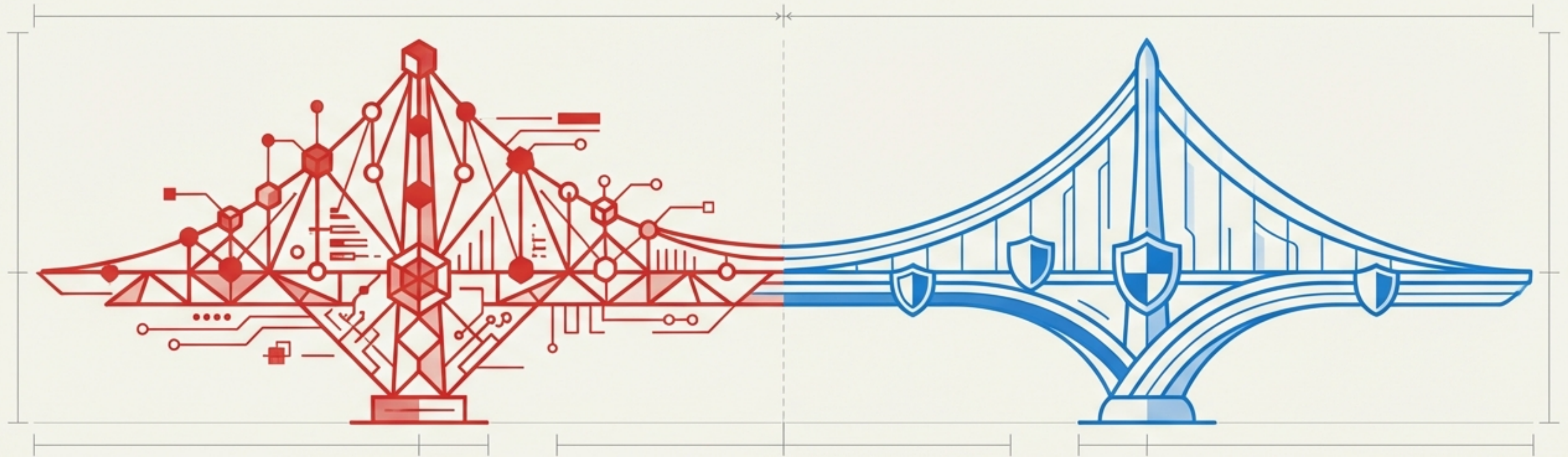
```
root/  
├── src/  
│   └── (Core Logic)  
├── main.py  
│   └── (Entry Point)  
└── Final_Security_Report.md  
    └── (Output Template)
```



Language: PYTHON

Architecture: Modular Pipeline

Bridging Offensive Testing & Defensive Strategy



SIMULATION

Successfully mimics modern password audit lifecycle.

SAFETY

Benchmarks resilience without compromising real user data.

INTELLIGENCE

Delivers clear entropy metrics and risk calculations.